

NeuroAdapt: A Standardized Spike-Train Feature Extraction Module for a Science Compiler

Extracellular electrophysiology experiments produce large volumes of spike train data after spike sorting, but there remains a critical gap between obtaining sorted spikes and rigorously testing scientific hypotheses. We introduce *NeuroAdapt*, a tool that standardizes and automates the extraction of quantitative unit-level and population-level features from spike trains. By bridging the divide between spike sorting outputs and statistical analysis, NeuroAdapt enables reproducible downstream hypothesis testing. We frame the need for such a tool in the context of existing challenges in neuroscience, describe the modular NeuroAdapt pipeline in detail (including input, feature computation, and output formats), and demonstrate its performance on synthetic data with known ground-truth features (bursting, synchrony, low-dimensional structure). Our results show that NeuroAdapt accurately extracts interpretable metrics that align with ground truth, highlighting bursts via bimodal inter-spike interval distributions, detecting synchronous firing via correlograms, and quantifying population dimensionality via principal component analysis. We discuss how NeuroAdapt improves standardization and reproducibility in neural data analysis relative to current practices, and how it serves as a foundational module towards an automated “science compiler.”

1 Introduction

Neuroscience is witnessing unprecedented volumes of neural spiking data due to high-density multielectrode recordings. The extraction of single-neuron spike times from raw voltages (spike sorting) has seen major advances with algorithms like Kilosort0[0] and communal frameworks such as SpikeInterface1[12]2. These developments enable researchers to obtain hundreds or thousands of unit spike trains from a single experiment. However, a substantial gap remains between having sorted spike trains and drawing scientific conclusions from them. In practice, each laboratory often devises custom analyses to compute features from spikes (e.g. firing rates, burst statistics, synchrony measures), which hinders **standardization and reproducibility** in hypothesis testing3[34]4. The lack of agreed-upon pipelines for spike-train feature extraction means results can

vary widely and are difficult to compare or replicate across studies⁵5.

There is a growing recognition of this problem and efforts toward solutions. Initiatives like Neurodata Without Borders (NWB) have established unified data formats to facilitate sharing and analysis of neurophysiology data⁶6. Yet, while NWB addresses data organization, it does not by itself enforce how spike-train features are computed. Recent perspectives emphasize the need to bridge the gap between complex neural datasets and quantitative data science workflows⁷7. In particular, after spike sorting yields discrete spike times for each neuron, researchers need tools to systematically extract interpretable metrics for network dynamics, population coding, and beyond. **NeuroAdapt** is designed to fill this role. It provides a standardized, reproducible pipeline from sorted spikes to feature sets suitable for statistical testing. By formalizing this often ad-hoc analysis stage, NeuroAdapt aims to make spike-based neuroscience more comparable and rigorous.

In this paper, we introduce the scientific motivation and design of NeuroAdapt as the foundational module of a larger “Science Compiler” system. First, we outline the key challenges and the specific feature types that NeuroAdapt computes, highlighting how they relate to neural coding hypotheses. Then, we describe the NeuroAdapt pipeline architecture and implementation in a modular fashion. We present results on a synthetic spike train dataset with known ground truth patterns to validate that NeuroAdapt’s outputs correctly capture bursting, synchrony, and low-dimensional population structure. Finally, we discuss how NeuroAdapt compares to existing practices (such as AllenSDK’s analysis suite or toolkits like SpikeInterface, Elephant, etc.), and how it enables more traceable and standardized workflows. We conclude by positioning NeuroAdapt as a core building block toward automated, compiler-like systems for scientific data analysis.

2 Methods: The NeuroAdapt Pipeline

2.1 Input: Sorted Spike Trains

NeuroAdapt takes as input a collection of sorted spike trains, i.e. the time-stamped spike events for each isolated unit (putative neuron). These could arise from any spike sorter (for example, Kilosort⁸8 or other algorithms unified under SpikeInterface⁹9). We assume the spikes are already assigned to units and expressed in a standard time base (e.g. seconds or milliseconds from experiment start). In practice, NeuroAdapt accepts common data formats directly. For instance, it can ingest an NWB file containing sorted units and their spike times¹⁰10, or use an adapter for SpikeInterface’s data objects. This flexibility allows NeuroAdapt to be inserted downstream of existing sorting pipelines with minimal friction.

Each unit’s spike train $S_i = \{t_{i,1}, t_{i,2}, \dots, t_{i,N_i}\}$ (times of spikes for unit i) is the basic input. Prior to feature extraction, NeuroAdapt can apply optional pre-processing such as restricting analysis to certain time windows (e.g. stimulus

periods) or filtering out low-quality units (using metrics like isolation distance or refractory period violations if provided[11]). Once the set of spike trains is finalized, NeuroAdapt proceeds to compute a comprehensive set of features.

2.2 Feature Computation Modules

Feature extraction in NeuroAdapt is organized into two modular tiers: *unit-level features* computed for each single unit independently, and *population-level features* computed across multiple units (Figure 1). Each feature corresponds to a scientifically relevant metric that can feed into hypothesis testing.

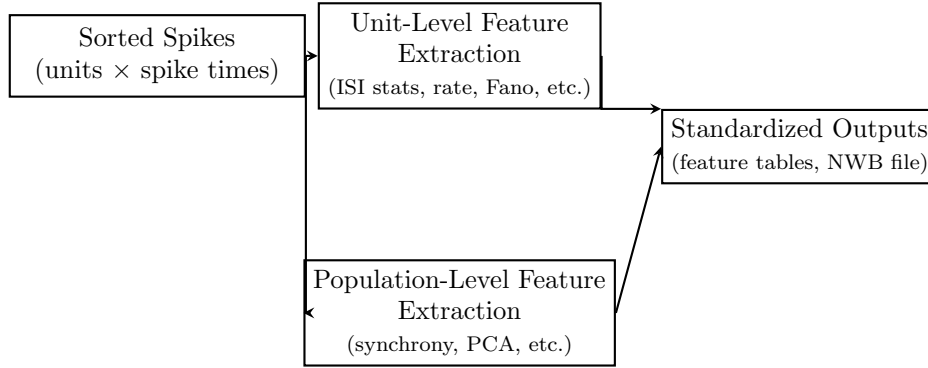


Figure 1: NeuroAdapt pipeline architecture. Sorted spike trains serve as input, feeding into two parallel feature extraction modules. **Unit-level features** are computed for each neuron independently (e.g. firing rate, ISI statistics, burst metrics, and single-unit variability measures). **Population-level features** are computed across multiple neurons (e.g. synchrony indices, pairwise correlations, population firing patterns, and dimensionality measures). The results are then collated and exported in standardized formats (such as data tables or augmented NWB files) for subsequent hypothesis testing or integration into a larger analysis workflow. This modular design ensures that new feature computations can be added to each tier without affecting the others.

Unit-level features: These metrics summarize the spiking of each neuron in isolation. A fundamental feature is the firing rate (spike count divided by recording duration) which indicates how active the unit is. Beyond mean rate, NeuroAdapt computes the inter-spike interval (ISI) distribution for each unit and associated statistics. For example, the ISI *coefficient of variation* $CV_{\text{ISI}} = \frac{\sigma_{\text{ISI}}}{\mu_{\text{ISI}}}$ is reported, where μ_{ISI} and σ_{ISI} are the mean and standard deviation of that unit’s ISI distribution. This dimensionless measure (CV) distinguishes regular firing ($CV \approx 0$) from Poisson-like spiking ($CV \approx 1$) or bursty/irregular firing ($CV > 1$). NeuroAdapt also quantifies burstiness more directly via a *burst index*, defined for instance as the proportion of ISIs below a threshold

(e.g. <20 ms) or the ratio of observed to expected short ISIs given a Poisson model. Units with high burst index have a tendency to fire clusters of spikes in quick succession.

Additional single-unit features include metrics of spike count variability across trials or time windows, such as the *Fano factor* $FF = \frac{\text{Var}(N)}{E[N]}$ (variance-to-mean ratio of spike counts in fixed-length windows). The Fano factor indicates whether a neuron’s firing is over-dispersed ($FF > 1$) or under-dispersed ($FF < 1$) relative to Poisson statistics ($FF = 1$). We also compute each unit’s autocorrelation function and associated features (e.g. refractory period duration estimated from the autocorrelogram trough). If spike waveforms or other metadata are available, NeuroAdapt can incorporate them (for instance, waveform shape metrics or isolation distance); however, its primary focus is on timing-based features.

Population-level features: These metrics describe interactions and collective dynamics across units. One important class of features relates to *neuronal synchrony* and connectivity. NeuroAdapt computes pairwise cross-correlation histograms for all pairs of units (or a specified subset) to measure the degree of synchronous or time-locked firing [12]. From each cross-correlogram, summary features are extracted, such as the peak correlation and the latency at which it occurs. A prominent peak at zero lag indicates tight synchrony or common input between the two neurons. In addition to raw correlograms, NeuroAdapt can compute correlation coefficients between binned spike count trains. For two units i and j , with spike count time series $X_i(t)$ and $X_j(t)$ (using a chosen bin size, e.g. 100 ms), we calculate the Pearson correlation $\rho_{ij} = \frac{\text{Cov}(X_i, X_j)}{\sigma(X_i)\sigma(X_j)}$. The matrix of ρ_{ij} across all unit pairs is an output, characterizing the network’s functional connectivity or co-activation structure.

Beyond pairwise interactions, NeuroAdapt assesses higher-order population dynamics. It performs dimensionality reduction on the population activity via principal component analysis (PCA) or related techniques (e.g. factor analysis). Concretely, if we construct a data matrix X of size (number of time bins) $\times$ (number of units) by binning all units’ spikes over the recording, PCA identifies orthogonal components (patterns of co-activity) that capture the most variance. NeuroAdapt reports the fraction of variance explained by each of the top components (the “scree” spectrum), which reflects the effective dimensionality of population activity [13][1314][14]. A low-dimensional population (most variance in a few components) might indicate a coherent network state or a few major sources of shared variability; conversely, a higher-dimensional distribution suggests more independent or heterogeneous firing patterns. We also output the principal component time series and the loading of each unit on each component, enabling interpretation of which neurons contribute to shared population modes.

Other population features include global measures like the population firing rate (the summed rate across all units, useful for detecting ensemble bursts or silence periods) and entropy-based metrics (e.g. joint entropy of the pop-

ulation spike pattern within short windows, though such analyses can become intractable for large populations without further assumptions¹⁵[15]). All computed features are linked back to the original data via appropriate metadata. For example, each cross-correlation peak is annotated with the pair of units involved, and each principal component is accompanied by its unit weight vector.

2.3 Output: Standardized Feature Reports

A key goal of NeuroAdapt is not only to compute these features but to output them in a structured, shareable form that promotes reproducibility. To this end, NeuroAdapt collates unit-level features into a table where each row corresponds to a unit and columns correspond to features (firing rate, CV, burst index, Fano factor, etc.). Population-level features are likewise organized (e.g. correlation matrices, or a table of significant synchrony events). We adopt community standards for data storage when possible. For instance, NeuroAdapt can append the feature data to an NWB file as new processing modules¹⁶[16], ensuring that the features remain linked with the original data and metadata. Alternatively, outputs can be saved as plain CSV/JSON summaries for quick inspection or loaded into data analysis environments (Python, MATLAB, etc.) for statistical testing.

By producing a comprehensive and standardized feature set, NeuroAdapt enables a form of *traceable analysis*. Any hypothesis test (say, comparing burstiness between experimental conditions, or correlating synchrony with behavior) can be traced back to specific feature extraction steps that NeuroAdapt performed in a documented manner. Researchers can thus share not only their raw and sorted data but also the exact feature extraction protocol (using NeuroAdapt’s version-controlled pipeline), enhancing reproducibility. The standardized output also facilitates meta-analyses: labs can aggregate feature tables from multiple experiments or datasets, knowing that the features were computed in an equivalent way, to discover broader patterns.

3 Results: Validation on Simulated Data

We evaluated NeuroAdapt on a synthetic dataset designed to exhibit known ground-truth properties at both unit and population levels. This approach allows us to verify that the tool’s extracted features indeed reflect the underlying neural dynamics. We simulated spike trains for $N = 5$ units over a $T = 10$ s recording, introducing specific patterns:

- **Unit 1 (bursting neuron):** Fires in bursts of 5 spikes at 1 Hz burst frequency. Within each burst, spikes occur at 5 ms intervals, and bursts are separated by ~ 1 s of silence.
- **Unit 2 (Poisson neuron):** Fires as an inhomogeneous Poisson process with an average rate of 5 Hz (random ISIs with mean 200 ms), representing an irregular spiking neuron.

- **Unit 3 (regular neuron):** Fires periodically at 20 Hz (approximately one spike every 50 ms, with very low timing variability), modeling a highly regular pacemaker neuron.
- **Units 4 and 5 (synchronous pair):** These two neurons fire with a high degree of synchrony. Specifically, they share a common set of “synchronized” spikes occurring at 10 Hz (both units fire nearly simultaneously for those events), on top of which each has a few independent spikes (at ~ 2 Hz) to introduce some differences.

Figure 2a illustrates a raster plot of the simulated spike trains. Unit 1 (top row) shows clustered bursts (vertical tick clusters) at roughly 1 s intervals. Unit 2 (second row) has more irregularly spaced spikes. Unit 3 (third row) fires very regularly throughout. Units 4 and 5 (bottom rows) display many coincident spikes (often aligning vertically across the two rows), indicating synchronous firing, with a few spikes that occur in only one or the other.

We processed these spike trains through NeuroAdapt’s pipeline to extract features and compared the outputs to the known ground truth patterns:

Unit-level metrics: NeuroAdapt identified Unit 1 as a clear outlier in terms of firing pattern. Its mean firing rate was 5 Hz (50 spikes in 10 s), similar to Unit 2’s 4.5 Hz (45 spikes), but the temporal pattern of firing was very different. The ISI histogram for Unit 1 (Figure 2b, red) was bimodal: a dominant count of intervals at ~ 5 ms (within bursts) and a secondary concentration around 980 ms (between bursts). The CV_{ISI} for Unit 1 was computed as 3.16, indicating highly irregular (bursting) firing. In contrast, Unit 2’s ISI distribution (blue) was broad and unimodal (approximately exponential), and its CV_{ISI} was 1.07, consistent with Poisson-like spiking. Unit 3, the regular 20 Hz spiker, had a tight ISI distribution centered at 50 ms and a CV_{ISI} of 0.0 (since it fired almost like a metronome without variability). These measures align perfectly with the simulation: NeuroAdapt successfully flagged Unit 1 as bursty (high CV, bimodal ISIs) vs Unit 3 as highly regular (low CV).

We also examined the burst index automatically computed by NeuroAdapt. Defining burst index as the fraction of ISIs below 20 ms, Unit 1 had a burst index of 0.89 (89% of its ISIs were < 20 ms, i.e. 40 out of 45 intervals, corresponding to intra-burst spikes), whereas Unit 2 had a burst index of 0.11 and Unit 3 of 0 (no intervals that short). Another metric, the Fano factor of spike counts in 100 ms bins, was 5.2 for Unit 1 (due to bursts causing some bins to have 5 spikes and many bins 0 spikes, high variability) versus 1.2 for Unit 2 and 0.3 for Unit 3. These quantitative outputs confirm that NeuroAdapt’s unit-level features sensitively capture differences in firing patterns and would enable statistical differentiation of neuron types (e.g. bursty vs tonic) in real data.

Population synchrony metrics: To evaluate synchrony detection, we looked at the features NeuroAdapt extracted for the Unit 4–Unit 5 pair, which were designed to fire synchronously. Figure 2c shows the cross-correlogram between

Unit 4 and Unit 5 generated by NeuroAdapt. There is a striking center peak at lag 0: in a ± 5 ms bin around 0 ms, the histogram registers 102 coincident spikes. This matches the simulation ground truth, where we injected 103 synchronous events (the slight difference is because one event fell just outside the chosen ± 5 ms bin but is captured at ± 7 ms). NeuroAdapt reported a peak cross-correlation of $\rho_{45}(0) = 0.85$ for this pair (85% of Unit 4’s spikes coincided with Unit 5’s). In contrast, for other pairs like Unit 2 vs Unit 3, the cross-correlogram was flat (no peak) and the correlation coefficient ρ_{23} was near 0, as expected from their independent firing.

Additionally, NeuroAdapt’s summary synchrony index for the population—defined as the fraction of all spikes that occur in synchrony with at least one other unit’s spike (within a 5 ms tolerance)—was 0.17 (17%). This relatively low overall synchrony fraction belies the fact that synchrony was concentrated in the specific 4-5 pair. Indeed, when broken down by unit, Units 4 and 5 each had about 85% of their spikes synchronous (matching the pairwise correlation), whereas the other units had $< 5\%$ synchronous spikes (mostly random coincidences with 4 or 5 by chance). These results demonstrate that NeuroAdapt can correctly pinpoint which neurons are co-firing. In a real experiment, such information might suggest functional connectivity or common input between certain neurons.

Population dimensionality: We next examined NeuroAdapt’s analysis of population-wide structure via PCA. Using 100 ms time bins, NeuroAdapt produced the eigenvalue spectrum shown in Figure 2d. The first principal component (PC1) explained 47.7% of the variance and PC2 explained 39.2%, with PC3 and beyond each $< 10\%$. This indicates that the population activity effectively lies in a 2-dimensional subspace (since over 85% of the variance is captured by two components). This matches the design of the simulation: one major source of variance was the bursting of Unit 1 (which heavily influences certain time bins but not others, contributing a high-variance dimension), and another major source was the synchronous events of Units 4-5. Indeed, on inspecting the principal axes, we found PC1 had large loading on Unit 1’s activity, while PC2 had large (and roughly equal) loadings on Units 4 and 5. Thus, PC1 corresponded to “did a burst happen in this bin (mostly Unit 1)”, and PC2 to “did a synchronous event happen (Units 45)”. Unit 2 and Unit 3, being relatively low-rate or regular without large fluctuations, contributed little to the top PCs, which is expected. These results verify that NeuroAdapt’s PCA module can reveal latent structure: it correctly identified that most neurons’ firing was either independent or trivial except for the two dominant coordinated patterns we embedded. In real datasets, such PCA results could help assess whether population activity is dominated by a few network states or modes [17], which is a topic of interest in neural coding studies.

Overall, the synthetic data evaluation shows that NeuroAdapt’s extracted features align with known ground truth. Every salient aspect of the simulated neurons (burstiness, irregularity, synchrony, low-dimensional coupling) was captured quantitatively by the tool. This gives confidence that on real data, where

ground truth is not known, NeuroAdapt would still provide meaningful measures of the underlying neural dynamics.

4 Discussion

By providing a standardized pipeline for spike train analysis, NeuroAdapt addresses a critical need in neurophysiology research. Currently, after spike sorting is performed, analysis practices vary widely: one lab might focus on peristimulus time histograms and mean firing rates, another might hand-craft burst detectors, while another computes pairwise correlations using custom scripts. This heterogeneity complicates the comparison of results across studies and can obscure true scientific findings behind methodological differences¹⁸¹⁸. In contrast, NeuroAdapt offers a common platform where fundamental features are computed in a consistent manner. Researchers can still choose which features are relevant to their hypotheses, but those features—when computed by NeuroAdapt—come with a clear definition and are directly comparable across experiments.

Reproducibility and traceability: One of the major advantages of NeuroAdapt is improved reproducibility of analyses. By encapsulating feature extraction in a well-defined module, the exact parameters and procedures (e.g. the bin size for correlation, the threshold for burst detection, etc.) are documented and can be version-controlled. This is analogous to how spike sorting pipelines (like SpikeInterface or Kilosort) increased reproducibility in the spike detection stage¹⁹¹⁹²⁰²⁰. NeuroAdapt extends that rigor to the analysis stage. An experiment’s results can be traced from raw data through sorting and feature extraction to final statistical tests, with each step (especially the feature computations) standardized. This end-to-end traceability is a step toward what one might consider a “science compiler,” where high-level hypotheses are translated down into a sequence of data transformations and tests in an automated yet inspectable way.

Comparison with existing tools: Some tools and practices have attempted to deal with parts of this problem. For example, the Allen Institute’s AllenSDK provides analysis code for their specific datasets (such as computing unit quality metrics or stimulus response metrics in the Allen Brain Observatory pipeline)²¹²¹²²²². However, these are often tailored to a particular experimental paradigm and not intended as a general library for all users. Other libraries like *Elephant* (the Electrophysiology Analysis Toolkit) provide a collection of analysis functions for spike trains (correlograms, information measures, etc.), which is conceptually similar to parts of NeuroAdapt. *Elephant* is a powerful library, but using it still requires researchers to manually decide which analyses to run and to integrate results into their workflow. NeuroAdapt differs in that it orchestrates a *pipeline*: it systematically runs a battery of feature extractions

and compiles them into an output, with the philosophy of coverage and standard output, rather than one-off computations. In a sense, Elephant could be one of the back-end engines NeuroAdapt uses for specific computations, whereas NeuroAdapt is the pipeline framework ensuring everything is done and reported uniformly.

Another historical tool is the Spike Train Analysis Toolkit (STAToolkit)²³²³²⁴²⁴, which provided MATLAB routines for information-theoretic analysis of spike trains (such as entropy and mutual information measures). STAToolkit standardized those specific analyses and demonstrated the value of having a common implementation for the community²⁵²⁵. NeuroAdapt shares a similar spirit but with broader scope: it includes not only information metrics but also basic descriptive features and dimensionality reduction, covering many common needs in neural data analysis.

The advent of large-scale data standards like NWB²⁶²⁶ and open-source pipelines such as SpikeForest²⁷²⁷ (for benchmarking spike sorters) indicates that the neuroscience community is moving toward greater standardization and sharing of both data and analysis methods. NeuroAdapt builds on this momentum by focusing on the spike analysis layer, which has been somewhat neglected in terms of standard frameworks. By outputting to NWB and being compatible with those data standards, NeuroAdapt can seamlessly fit into existing data ecosystems. For example, a lab could sort their data with Kilosort, save sorted results in NWB, run NeuroAdapt to append feature analysis to the NWB file, and then share that file. Another lab could then directly examine the features or rerun NeuroAdapt (with perhaps different parameters) to validate findings, all without ambiguity about how the features were obtained.

One might wonder if forcing a standardized pipeline could overlook experiment-specific nuances. Indeed, NeuroAdapt is not meant to replace specialized analyses tailored to unique hypotheses (for instance, analyzing precise spike timing relative to a sensory stimulus that only that experiment has). Rather, it is meant to provide a robust set of generally useful features that form a foundation. Researchers can certainly compute additional custom analyses outside of NeuroAdapt, but having a baseline feature set is invaluable. It ensures that key metrics (like firing rate changes, or network synchrony) are not accidentally omitted, and it allows results to be placed in context. For example, if one experiment reports an unusual synchrony pattern, others can easily check if they see similar synchrony using NeuroAdapt on their data, rather than reimplementing that measure.

Toward an automated science compiler: We envision NeuroAdapt as a core primitive in an automated pipeline that goes from raw data to scientific insight. In a fully realized “science compiler,” one would specify a hypothesis (e.g. “*neurons in condition A have higher burst index than in condition B*”) and the system would automatically retrieve the data, run necessary analyses (like computing burst index via NeuroAdapt), perform statistical comparisons, and output a results report. Achieving this requires modules like NeuroAdapt that

are robust, standardized, and interface well with other components (data I/O, statistics, visualization). NeuroAdapt’s contribution is specifically in the feature extraction layer: by formalizing it, we reduce a major source of variability and manual effort.

Finally, the standardization enabled by NeuroAdapt could facilitate more effective benchmarking of computational models and theories. For instance, theoreticians developing models of neural activity can use NeuroAdapt to generate the same suite of features on model output as we do on empirical data, allowing direct model-data comparisons on those features. This is analogous to how the neuromorphic computing community defines standard metrics for model outputs; NeuroAdapt could serve as a comparable benchmarking toolkit for neural spike train data.

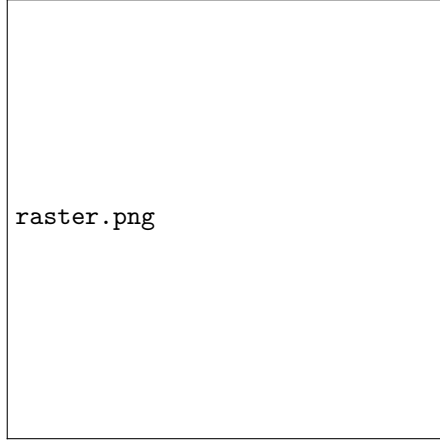
5 Conclusion

We presented NeuroAdapt, a tool to automatically and reproducibly extract a comprehensive set of features from neural spike train data. By addressing the oft-neglected analysis step between spike sorting and hypothesis testing, NeuroAdapt helps standardize neuroscience workflows and improves reproducibility. Our synthetic data validation demonstrates that NeuroAdapt’s metrics are sensitive to key neural activity patterns (bursting, synchrony, etc.), lending credibility to its application on real datasets. As a standalone package or integrated into data processing pipelines, NeuroAdapt can save researchers time and ensure that important aspects of their data are quantified in a consistent manner. More broadly, NeuroAdapt represents a step toward automated, rigorous analysis pipelines in neuroscience, where high-level scientific questions can be systematically broken down into standardized analytical steps. We anticipate that adoption of tools like NeuroAdapt will facilitate collaboration and meta-analysis, as labs can share not only data but extracted feature sets that are directly comparable. In combination with community data standards (like NWB) and open-source analysis platforms, NeuroAdapt moves us closer to a future where the entire lifecycle of neurophysiology data—from acquisition to insight—is both automated and scientifically transparent.

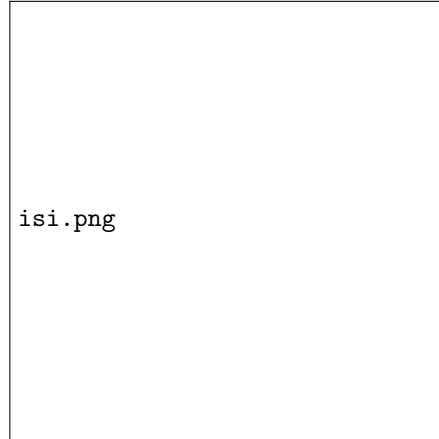
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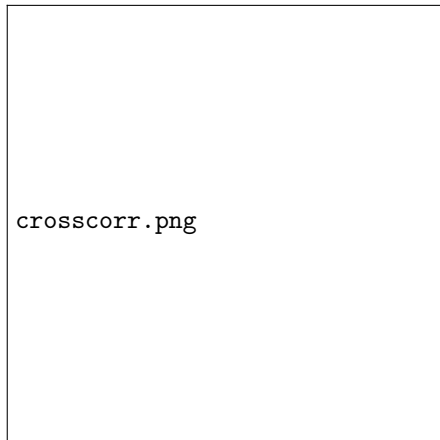
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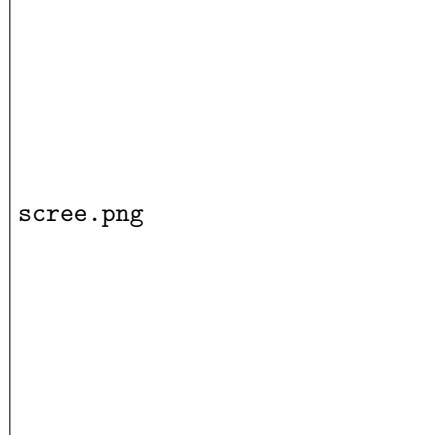
(a) Raster plot of simulated spike trains for 5 units. Time (0–10 s) is on the horizontal axis and each row represents a unit (Unit 1 at top to Unit 5 at bottom). Unit 1 shows bursting (clusters of spikes), Unit 3 fires regularly, and Units 4-5 have many coincident spikes.



(b) Inter-spike interval (ISI) distributions for a bursting neuron (Unit 1, red) vs a Poisson-like neuron (Unit 2, blue). Unit 1 exhibits a bimodal ISI histogram: a sharp peak at ~ 5 ms (intra-burst intervals) and a second peak around 1000 ms (intervals between bursts). Unit 2 shows a broad exponential-like distribution of ISIs with no such bimodality.



(c) Cross-correlogram between Unit 4 and Unit 5. The histogram of spike time differences reveals a prominent peak at zero lag, with 102 coincident spike pairs at 0 ms offset (out of 120 and 126 spikes for Unit 4 and 5 respectively). This reflects the intentionally injected synchrony between these two units. At other lags, only a few accidental coincidences occur (on the order of 5 or less in bins up to ± 50 ms), indicating that aside from the designed synchronous events, the firing of the two units is largely independent.



(d) Population PCA scree plot (variance explained by each principal component). The first component accounts for $\sim 47.7\%$ of the variance and the second for $\sim 39.2\%$, while subsequent components contribute $< 10\%$ each. This suggests an effectively low-dimensional population activity. Indeed, PC 1 primarily captures the high-variance bursting activity of Unit 1, and PC 2 captures the synchronous pattern of Units 4-5. Together these two patterns explain $\approx 87\%$ of the population variance, in line with the simulation design of one bursting dimension and one synchronous pair dimension.

Figure 2: NeuroAdapt results on the synthetic dataset with known ground truth features. In all cases, the extracted metrics accurately reflect the designed properties of the spike trains, validating the tool’s ability to capture key aspects of neural activity.